

Ultrasound Image Enhancement Pipeline

Speckle Despeckling and CLAHE Feature Enhancement
for Breast Ultrasound Segmentation Preprocessing

Why Ultrasound Images Need Enhancement

- Ultrasound is the most widely used non-invasive diagnostic imaging modality
- But raw B-mode frames have two persistent problems:
 - Low contrast — histograms are skewed toward low intensities
 - Multiplicative speckle noise — degrades fine detail, edges, and lesion boundaries
- Filtering out speckle on its own induces blurring and destroys structure that matters for diagnosis
- Goal: suppress speckle, then enhance features — in that order

$$I(x,y) = J(x,y) \times N(x,y)$$

Observed image = clean tissue signal $\times$ multiplicative speckle field

Speckle is not additive Gaussian noise — it multiplies the underlying signal, so simple subtraction-based filters do not remove it cleanly.

Two-Stage Processing Pipeline



Stage 1 removes multiplicative speckle noise; Stage 2 recovers and enhances local contrast and edges that speckle filtering tends to blur.

This notebook pipeline outputs an enhanced single-modality image feeding directly into a segmentation backbone (see slide 12).

Stage 1 — Speckle Despeckling

Speckle model: $I = J \times N$ (N = multiplicative speckle field to be estimated / removed)

Classical — Lee Filter (default)

- Adaptive local-statistics filter
- Local coefficient of variation: $C_i = \sigma / \mu$, within a sliding window
- Homogeneous regions (low C_i) → smoothed toward the local mean
- Feature / edge regions (high C_i) → left mostly unchanged
- No training data required — usable immediately

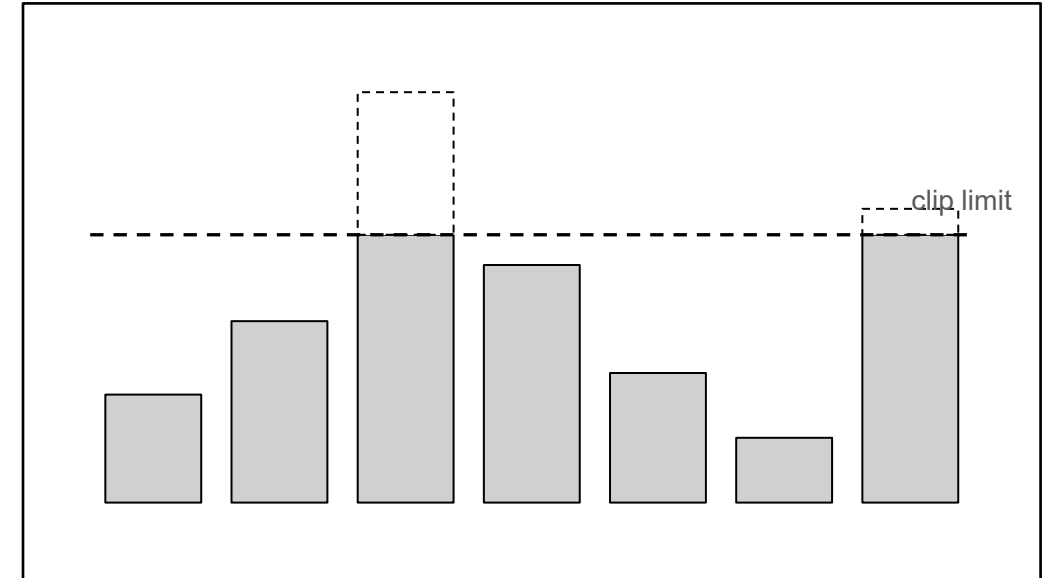
CNN-Based (paper's method, optional)

- 3-layer network: Conv+ReLU → Conv+ReLU → Conv+BN+Tanh
- Learns to estimate the multiplicative speckle field N
- Despeckled image = I / N
- Trained on synthetically generated speckle pairs (no real paired data needed)
- Available as an alternative once trained

Stage 2 — Why CLAHE, Not Plain Equalization

- Global histogram equalization stretches contrast using the whole image — fails when the histogram is skewed, as ultrasound typically is
- Adaptive Histogram Equalization (AHE) equalizes small tiles independently — but over-amplifies speckle noise in flat regions
- CLAHE clips each tile's histogram at a limit before equalizing — local contrast is enhanced without letting noise explode

Tile histogram (schematic)



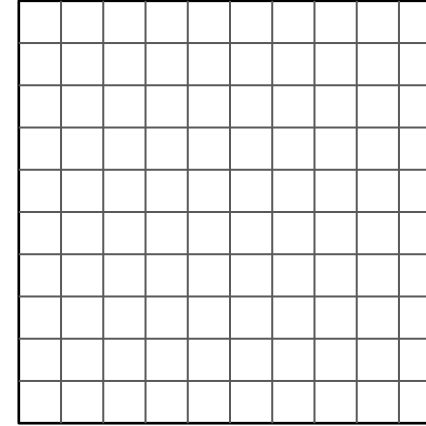
Excess above the clip limit is redistributed across the tile's histogram bins before the transform is computed.

Automatic Tile Sizing via Image Entropy

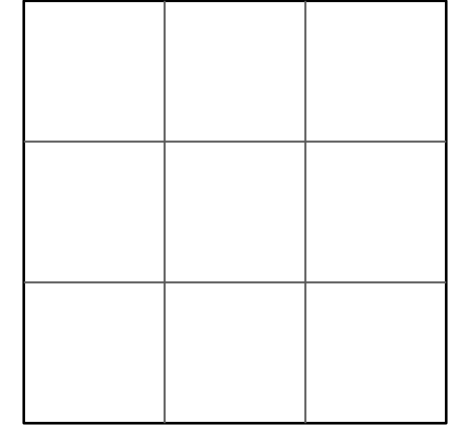
$$E = - \sum p_k \log_2(p_k), \quad k = 0..255$$

$$s = (M - E_{\max}) e^{(-\lambda E)} + E_{\max}$$

- High entropy (detailed, textured frame) → small tiles (~8×8 px) for fine local adaptation
- Low entropy (near-uniform frame) → tiles up to M/2, avoiding unnecessary local processing
- $\lambda = 0.7$, $E_{\max} = 8$ bits — pipeline defaults



High entropy →
fine tile grid



Low entropy →
coarse tile grid

Target Distributions and Interpolation Kernels

3 Target Histogram Shapes (per tile)

- Uniform — classic histogram equalization
- Rayleigh — $\text{out} = \alpha \sqrt{2 \ln(1/(1-\text{cdf}))}$
- Exponential — $\text{out} = -\alpha \ln(1-\text{cdf})$
- Clip limit controls aggressiveness: too high $\rightarrow$ oversaturated; too low $\rightarrow$ flat

3 Interpolation Kernels (blend tiles)

- Bilinear — nearest 2×2 tiles, linear weights
- Cubic — 4×4 tile neighborhood, smoother transitions
- Lanczos-3 — 6×6 tile neighborhood, sharper detail preservation
- 3 distributions × 3 kernels = 9 combinations evaluated

Pipeline default — Rayleigh distribution + Lanczos-3 interpolation, clip limit ≈ 0.01 (see next slide for why)

Why Rayleigh + Lanczos-3 Performs Best

- Rayleigh statistics naturally model the amplitude distribution of fully-developed ultrasound speckle — matching the target shape to the tissue's underlying physics avoids an artificial-looking contrast stretch
- A low clip limit (≈ 0.01) keeps the transform close to identity in flat regions, enhancing contrast mainly where local structure already exists
- Lanczos-3's wider support and sharper roll-off blends neighboring tile transforms with less block artifact than bilinear or cubic interpolation
- Confirmed by both radiologist subjective scoring (highest among 9 combinations) and objective SSIM / EPI / UQI metrics

Measuring Enhancement Quality

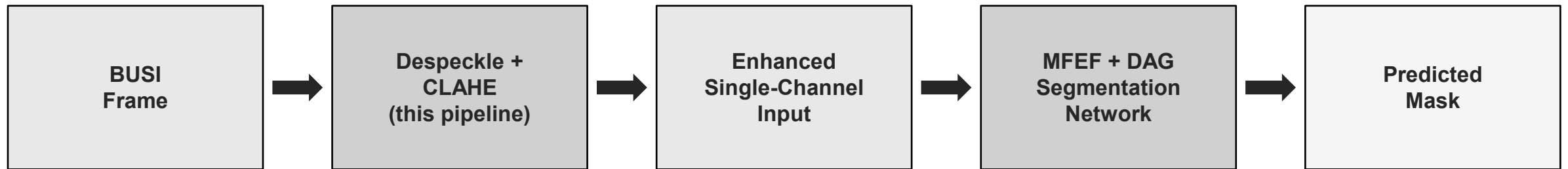
Metric	What It Measures	Good Value
PSNR	Pixel-level fidelity vs. a reference frame (decibels)	Higher is better
SSIM	Structural similarity — luminance, contrast, structure	Close to 1
EPI	Edge preservation index — how well edges survive processing	Close to 1
UQI	Combined contrast, luminance, and correlation distortion	Close to 1

All four metrics are used in the comparative analysis (Section 7 of the accompanying notebook) to score the 9 CLAHE distribution × interpolation combinations.

Dataset — BUSI (Breast Ultrasound Images)

- ~780 breast ultrasound images from 600 female patients, collected in 2018 at Baheya Hospital, Cairo, Egypt
- 3 classes: normal, benign, malignant — each image paired with an expert-annotated ground-truth segmentation mask
- A widely used public benchmark for breast ultrasound segmentation research
- Why used here:
 - Ground-truth masks are needed to train and evaluate the downstream segmentation network
 - Realistic clinical speckle and contrast characteristics — a genuine testbed for this preprocessing pipeline
 - Known limitation: prior audits report image/annotation quality issues in a meaningful share of the set (roughly 40%), so a cleaning pass precedes use

Where This Fits — Preprocessing for AMFU-Net



- Replaces the original dual-modality fusion stage of AMFU-Net with single-modality despeckling + CLAHE preprocessing
- Feeds directly into the unchanged MFEF (multi-scale feature extraction and fusion) + DAG segmentation backbone

Summary

- Two-stage pipeline: despeckle (Lee filter / CNN) → CLAHE (entropy tile sizing + Rayleigh distribution + Lanczos-3 interpolation)
- Reduces speckle noise while enhancing diagnostically relevant contrast and edges
- Validated end-to-end on synthetic data; BUSI selected for real-data evaluation and downstream segmentation
- Feeds directly into the AMFU-Net-based segmentation pipeline as the single-modality preprocessing stage
- Next steps: clean the BUSI subset, run the 9-combination comparative analysis on real frames, and integration-test with the MFEF + DAG backbone